Lecture 4: Fundamentals of Machine Learning Pt. 1

INFO 1998: Introduction to Machine Learning

CDS Education

Mid Semester Project Check-In 👀

- Starting next week (can complete anytime from 10/07 - 10/18)!

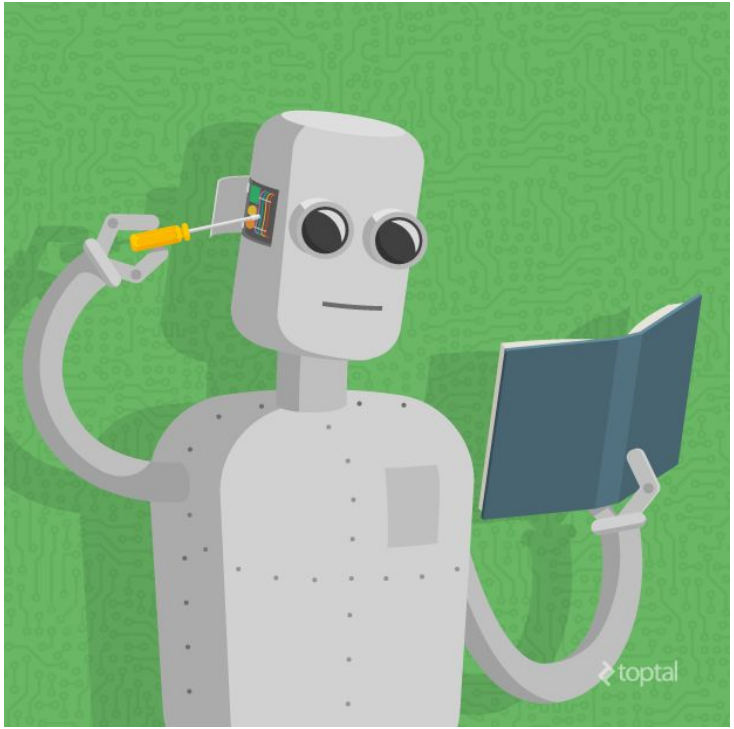
- Attend OH, and fill out a Google Form (will be posted on EdStem)
  - Expecting hypothesis/question/problem to solve
  - Chosen dataset
  - Some progress on data cleaning/data visualization

Agenda

Today's Goal: be able to write code to do some kind of ML (to some extent)

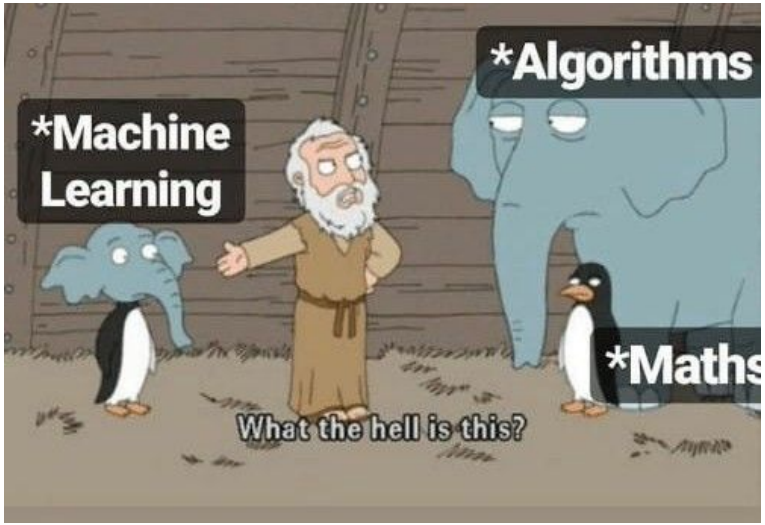
- **Define Machine Learning:** or like, 5 definitions
- **Start learning the language of ML:** There's a lot of terminology!
- **Try Linear Regression (via ScikitLearn)**: Our first ML algorithm!
- **Introduce our Workflow**: An outline for developing an ML model
- **Discuss Some Important Considerations**: What should we be thinking about as we're MLing?

What's Machine Learning? Part 1: what does an ML engineer do



Some Definitions of ML

- Give computers the ability to learn without being explicitly programmed
- Build a useful mathematical model, based on sample data, to make inferences
- Take in data and make predictions or decisions
- Help your computer learn patterns



Machine Learning can involve:

- Preprocessing data
- Splitting and selecting pieces of data
- Doing mathematical analysis on the data
- Deciding what data structures are needed to efficiently implement algorithms
- Implementing accuracy metrics
- ...and a lot more

What we're gonna do:

Our main tasks:

- **Formulate** a problem
- Find and **understand** data for that problem
- Choose a specific **algorithm class** to **solve** the problem
- Choose different parts of the data to **best** solve the problem
- Find which pandas, numpy, and scikit-learn functions do what we want
- Interpret the results and **fine-tune** our model

Write as little code as possible!

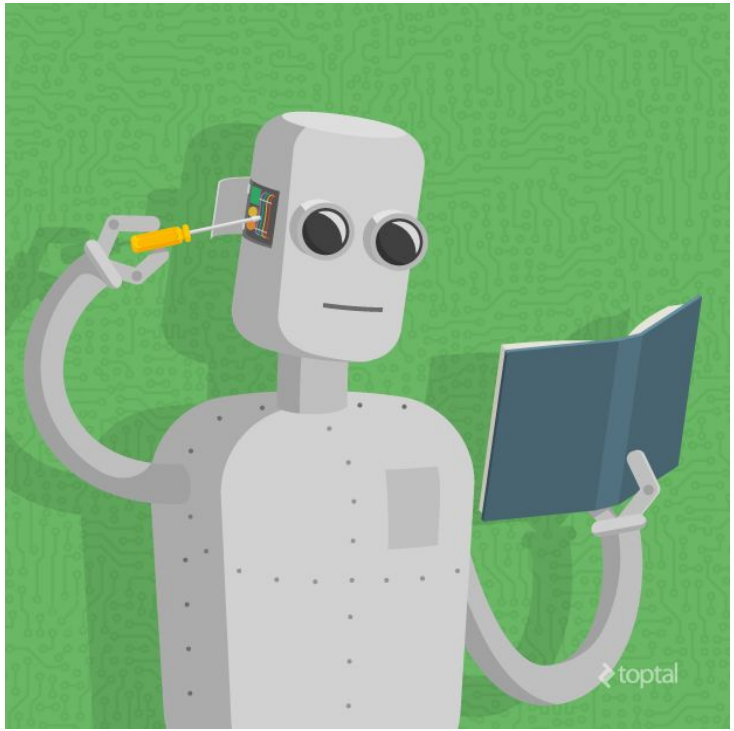
- Use pandas to deal with data
- Use numpy to do math
- Use scikit-learn ("sklearn") to make & analyze ML models

Quick analogy: studying

- Setup
  - Goal: Be able to solve the test problems *well*
  - Resources: Practice problems + answers
- Method
  - You study those practice problems and answers. Given a problem, how do you get the answer?
- Result:
  - On the real test, the problems aren't the exact same as the practice problems. But they're similar!
  - Since you learned generally how to solve the practice problems, you can solve the similar test problems too :)

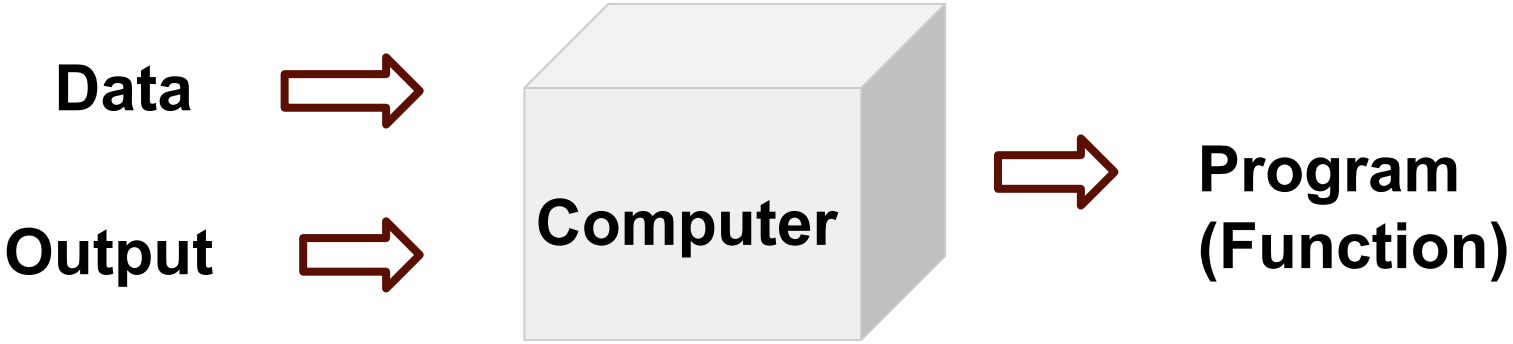
*Note: this analogy describes supervised learning

What's Machine Learning? Part 2: like seriously what is it

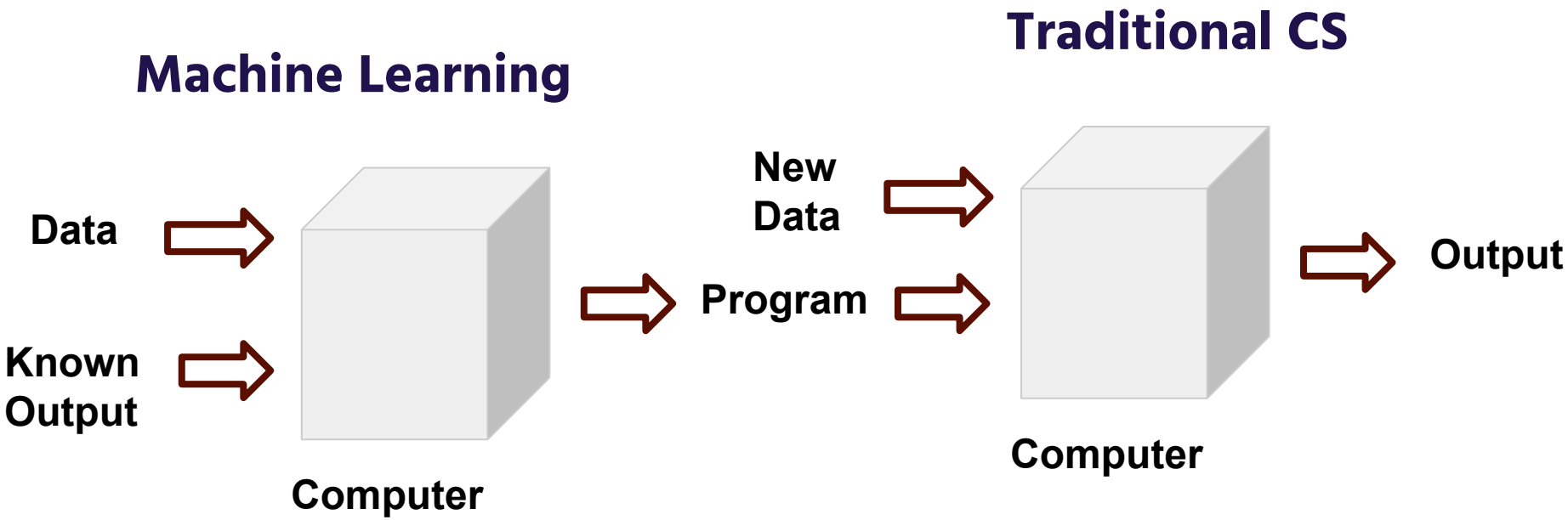


Traditional Computer Science

Machine Learning



Using Machine Learning



Physics: A White-Box View of the World

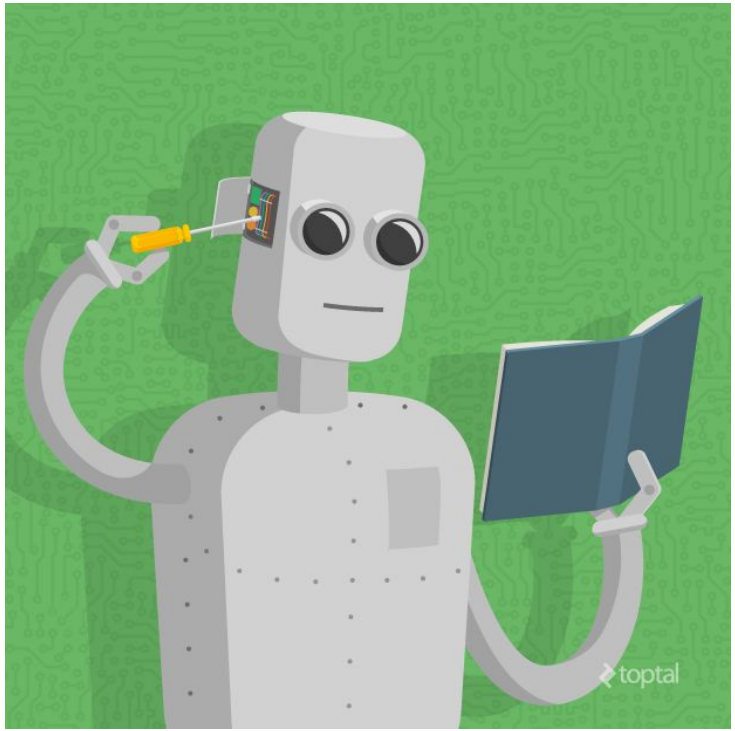
- We have studied the world and crafted **models** *by ourselves* to represent it
- Our models are interpretable
- **White-box algorithms:** The inner workings of the algorithm are transparent

$$F = ma$$

Black Box Models

- Perhaps we can derive a process to solve the problem
  - Determine what a function would roughly look like
  - Think of relevant inputs
  - Allow the function to *build itself*
- **Black Box Model:** Results may not be interpretable!

What's Machine Learning? Part 3: what's a model?



What’s a model?

- The output of a machine learning algorithm
- A procedure to produce some outputs when given some inputs
- A relationship between inputs and outputs
- A guess at how inputs and outputs are related
- A set of assumptions we’re imposing on the dataset
- A parametrized function we can configure

Review: Dataset Structure

- Rows are data points
  - AKA samples
- Columns are features
  - A sample is made of lots of features, including the goal

	Name	Age	Major
0	Ann	19	Computer Science
1	Chris	20	Sociology
2	Dylan	19	Computer Science
3	Camilo	NaN	NaN
4	Tanmay	NaN	NaN

A Sample (Regression) Task

name	city	state	adm_rate	undergrads	cost	compl_4	median_hh_inc	median_earnings
Cornell University	Ithaca	NY	0.1507	14226	63596	0.8639	80346.48	73600
Washington University in St Louis	Saint Louis	MO	0.1674	7032	65887	0.8643	79298.58	66300
Lafayette College	Easton	PA	0.3025	2505	61905	0.8653	85923.51	67500
Johns Hopkins University	Baltimore	MD	0.1412	5862	63509	0.869	81539.46	69800
Vanderbilt University	Nashville	TN	0.1168	6857	62320	0.8697	76279.78	64500

What are some things we can do?

name	city	state	adm_rate	undergrads	cost	compl_4	median_hh_inc	median_earnings
Cornell University	Ithaca	NY	0.1507	14226	63596	0.8639	80346.48	73600
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Predict Median Graduate Earnings

name	city	state	adm_rate	undergrads	cost	compl_4	median_hh_inc	median_earnings
Cornell University	Ithaca	NY	0.1507	14226	63596	0.8639	80346.48	73600
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Pick Some Features

name	city	state	adm_rate	undergrads	cost	compl_4	median_hh_inc	median_earnings
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Our Goal?

name	city	state	adm_rate	undergrads	cost	compl_4	median_hh_inc	median_earnings
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Vanderbilt University	Nashville	TN	0.1168	6857	62320	0.8697	76279.78	64500
Rutgers University	New Brunswick	NJ	0.5845	35102	29076	0.5838	82669.68	?
Case Western Reserve University	Cleveland	OH	0.3627	5039	59467	0.6311	69873.4	?

ML Algorithms

- We pick different kinds of algorithms to accomplish different tasks

- Classification
  - Group Data Into Distinct Classes

- Regression
  - Based on an input, provide a continuous-value output

- **“All Models Make Assumptions”**

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Attendance!

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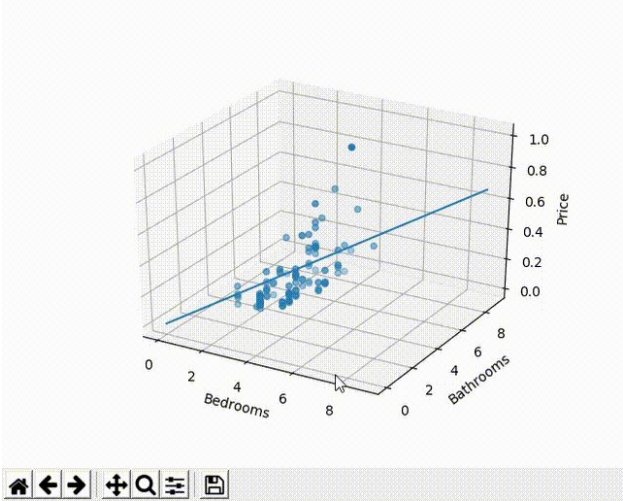
Linear Regression

Our first ML model!

Linear Regression

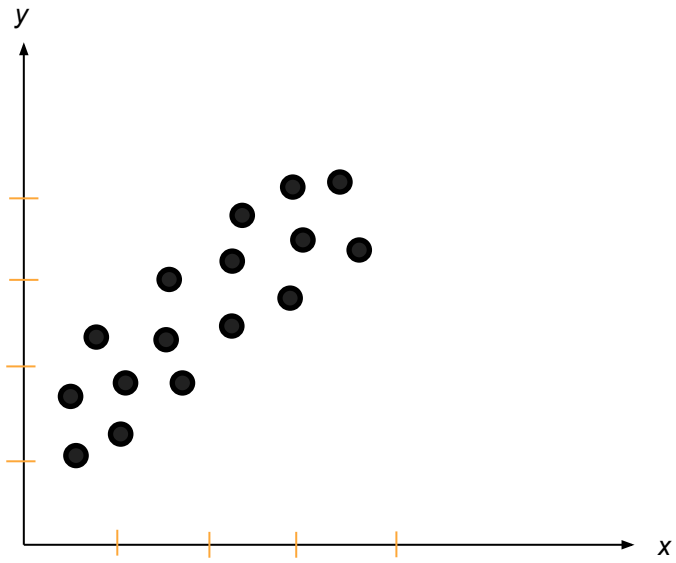
$$y = B_0 + B_1 x_1 + \ldots + B_p x_p + \varepsilon$$

- x is an input; $x_1, x_2, \ldots, x_p$ are the features of x
- y is an output (usually a single value)
- B's are "weights"
  - a linear regression equation is defined by its B's
  - "program" produced by ML
- Given a set of x's and y's, find the "B"-s that "most closely" satisfies the equation above for training points
- Plug in *new* values to of x to predict respective y



$y = B_0 + B_1x_1 + \ldots$ is a model

- A relationship between inputs and outputs
  $y = B_0 + B_1x_1 + \ldots$ relates inputs to outputs
- A guess at how inputs and outputs are related
  but $y = B_0 + B_1x_1 + \ldots$ is just a guess/estimate; it's not exactly true
- A set of assumptions we're imposing on the dataset
  We're assuming output is linearly related to input features
- A configurable thing (hyperparameters)
  Sorry, we don't cover very much linear regression configuration here 😅



$y = B_0 + B_1x$

$y = B_0 + B_1x_1 + \ldots$ is a model

Use algorithm to “learn” parameters that give us this line of best fit

- **A relationship between inputs and outputs**
  $y = B_0 + B_1x_1 + \ldots$ relates inputs to outputs
- **A guess at how inputs and outputs are related**
  but $y = B_0 + B_1x_1 + \ldots$ is just a guess/estimate; it’s not exactly true
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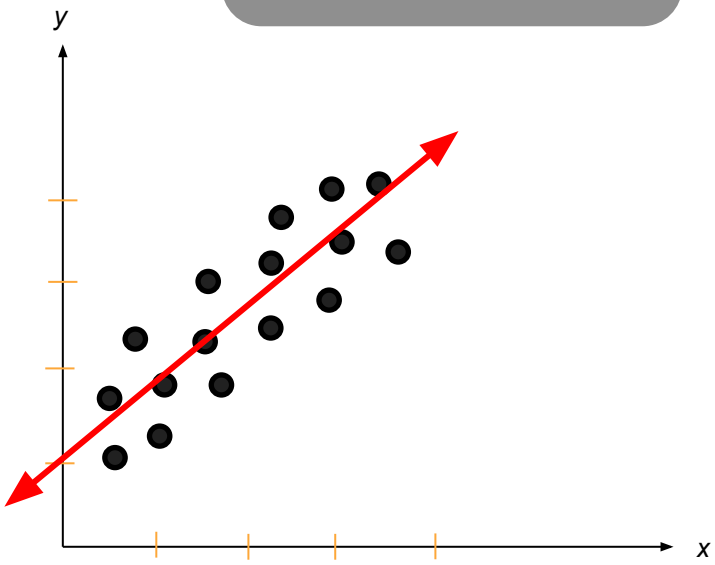
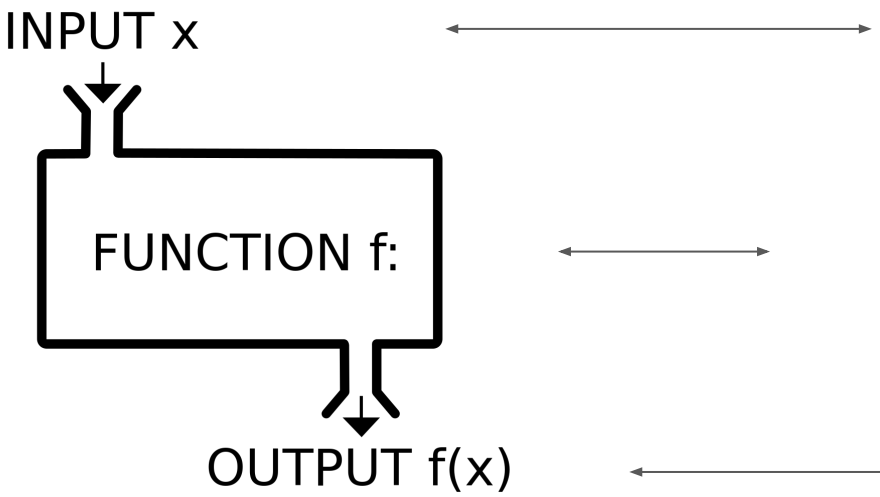


$y = B_0 + B_1x$

Linear Regression

Function



$x_1 \ldots x_p$

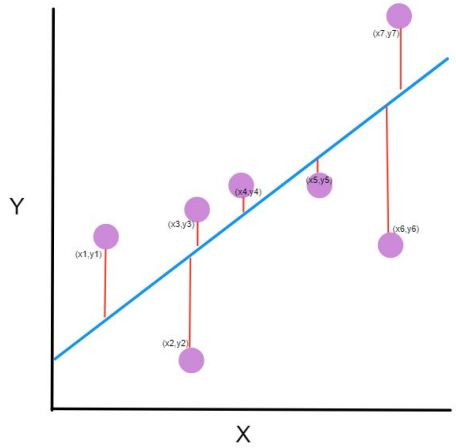
$y = B_0 + B_1 x_1 + \ldots + B_p x_p$

y

Linear Regression: Ordinary Least Squares

- *Ordinary least squares* method of linear regression calculates the weight vector B by minimizing the **mean-squared error** of the predicted y-values

- Other types of linear regression (i.e. ridge regression) use different *loss functions* to calculate the weights

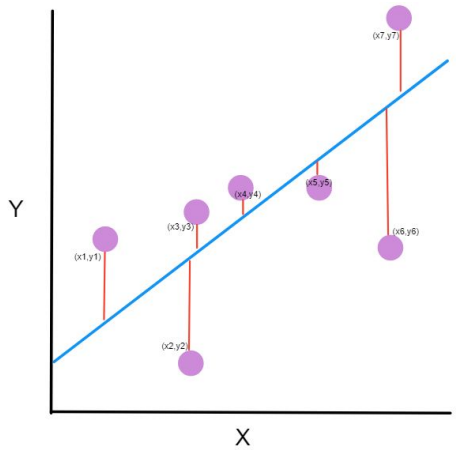


Mean

Error Squared

$$\mathrm{MSE} = \boxed{\frac{1}{n}\sum_{i=1}^{n}}\left(\boxed{Y_i - \hat{Y}_i}\right)^{\boxed{2}}$$

“Training” a Model



- Dataset of n training points
- Datapoints: (X, Y_i) -> (input, output)
- Objective: Minimize MSE

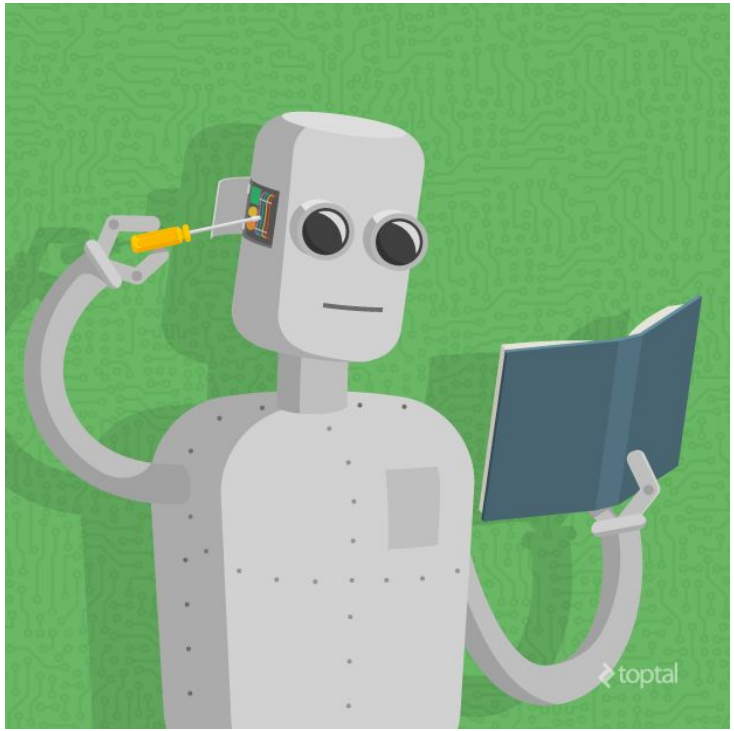
1. Use the x values in our dataset to make a prediction
   a. Note: x is a vector
2. Compare our prediction to the real Y_i
3. *Update B to get a better prediction*
   a. Special Algorithm: Gradient Descent
4. Repeat until MSE is as small as possible

$$\hat{Y}_i = B_0 + B_1 x_1 + \ldots + B_p x_p + \varepsilon$$

Mean Error Squared

$$\text{MSE} = \boxed{\frac{1}{n}\sum_{i=1}^{n}}\boxed{(Y_i - \hat{Y}_i)}^{\boxed{2}}$$

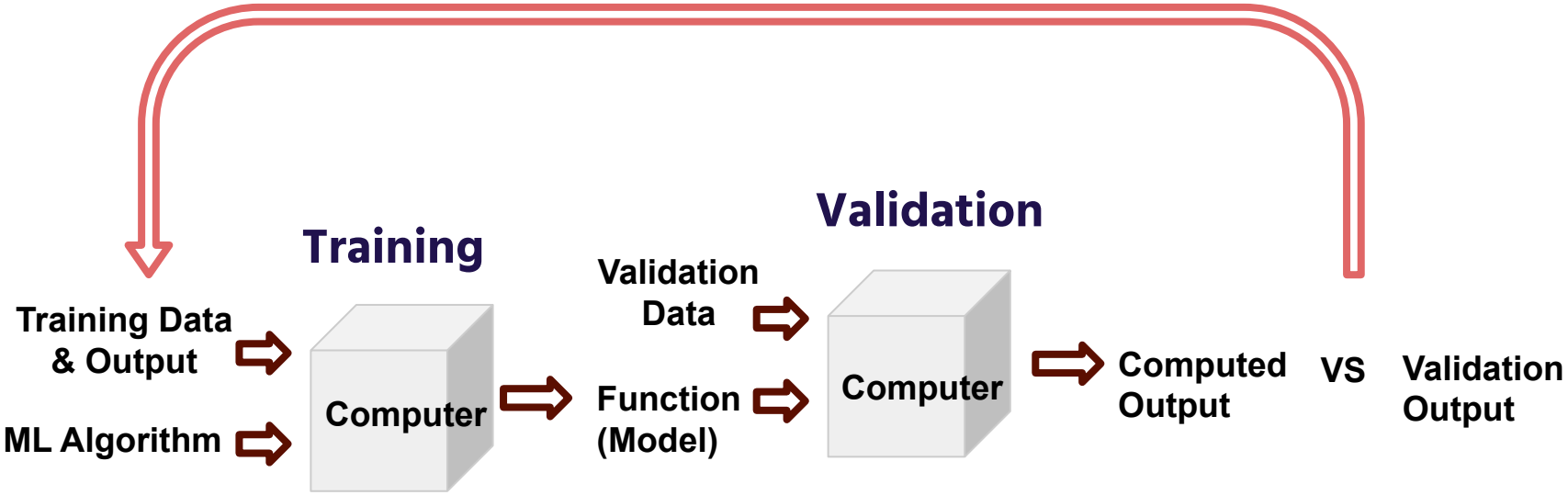
What's Machine Learning? Part 4: What makes a *good* model?

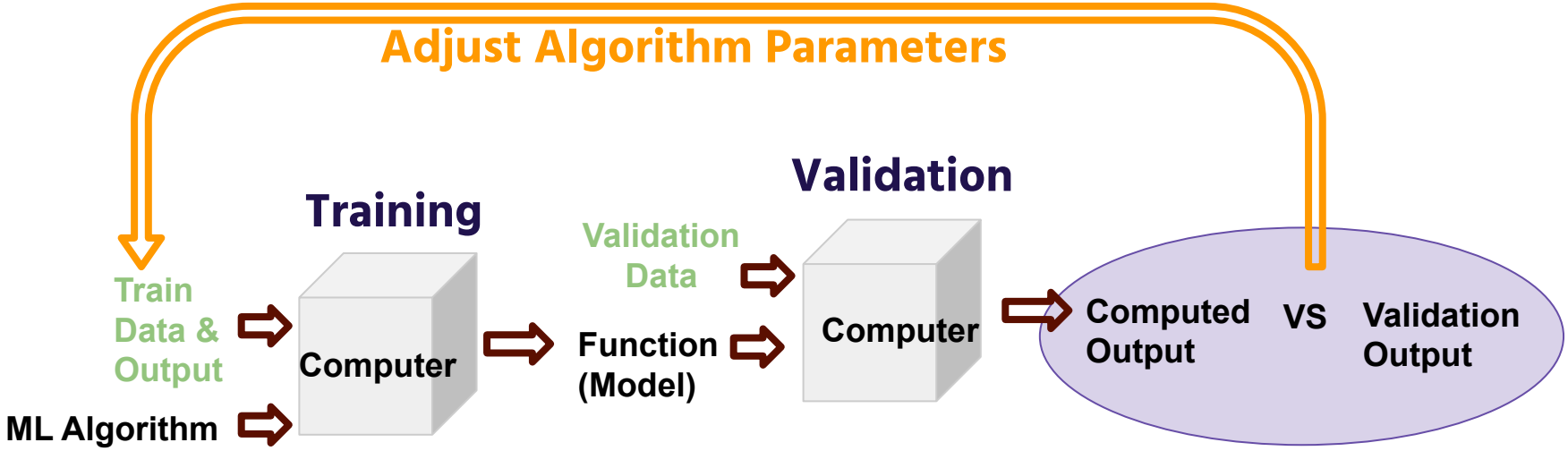


Terminology: Training and Validating

- Split data into two sets
- Train model on one, validate on the other
- “Model training” = learn a relationship/program
  - e.g. give the linear regression data so it can define the B’s
- “Model validation” = see if the learned relationship is accurate on other data

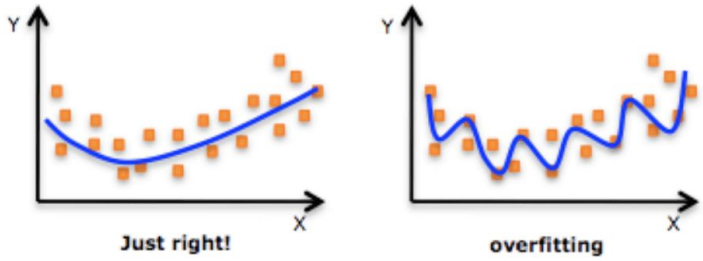
Our ML Workflow





1. Select data
2. Assess model accuracy
3. Adjust Model

Pitfall of Training: Overfitting



Model is accurate for **train** data

$\neq$

Model can accurately predict **new** data

- We learned the specific mapping from **train input to train outputs…**
- But, we didn't learn the data's **general patterns** 🥲🥲🥲🥲🥲🥲

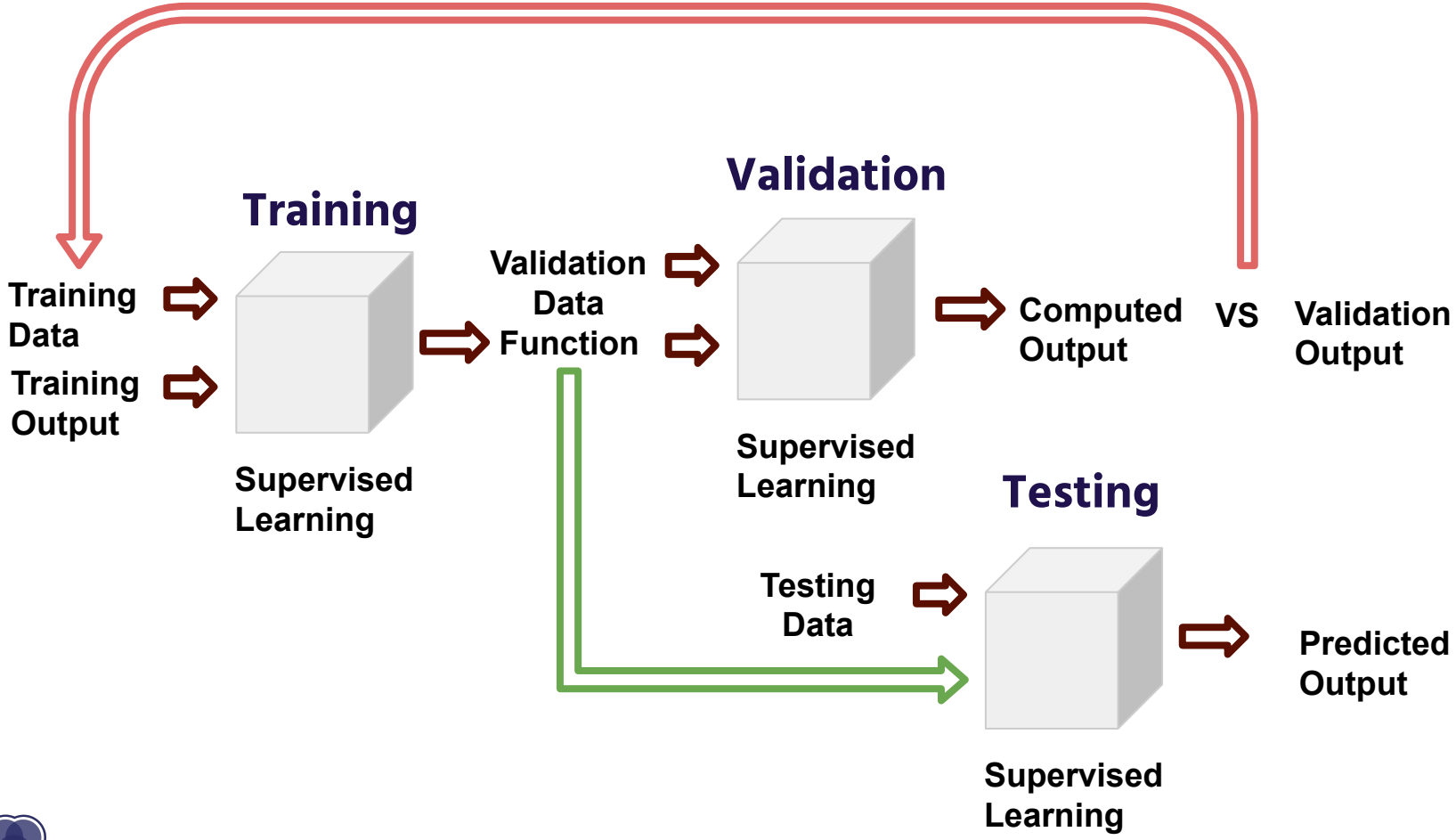
Solution: train on part of data, and check accuracy on a separate part of data (*validation* set)

Pitfall of Validation: Overfitting

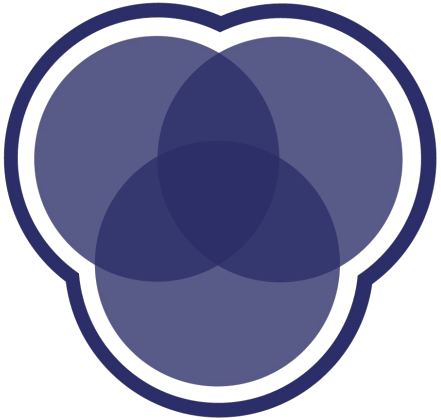
Predicting well on validation set $\neq$ Predicting well on new data

- We used the validation set to make our adjustments.
  - ⇒ Our model is **biased** to the validation set. 🥲🥲

Solution: keep a separate, rarely-used ***test*** set


Training
Training Data
Training Output
Supervised Learning
Validation Data Function
Validation
Supervised Learning
Computed Output
VS
Validation Output
Testing
Testing Data
Supervised Learning
Predicted Output

Demo



Model Goals

When training a model we want our models to:

- Capture the trends of the training data
- Generalize well to other samples of the population
- Be moderately interpretable

The first two are especially difficult to do simultaneously!
The more sensitive the model, the less generalizable and vice versa.

Things to Keep In Mind

- Linear Regression is just one algorithm — we'll cover many more! 🤠
- The "model" produced by an algorithm is not always a simple equation like in linear regression.
- Validation is *really* important.
  - Overfitting is a huge problem!
  - We'll delve deeper in the next few lectures...

Coming Up

Assignment 3: Due tonight at 11:59pm EST

Assignment 4: Due at 11:59pm EST on **October 9th**

Next Lecture: Assessing Model Accuracy **+** Fundamentals of ML

(a.k.a. *What's Machine Learning? Part* ∞)